

Chapitre 9 : Interfaces utilisateurs

Solution des exercices

Exercice 1

```
from tkinter import *

def convertir():
    fah = float(EntreePouce.get()) * 1.8 + 32
    txtCm.configure( text = str(fah))

fen1 = Tk()

label1 = Label(fen1, text = "Temperature en celsius:")
label1.grid(row = 0, column = 0, sticky = E)

EntreePouce = Entry(fen1)
EntreePouce.grid(row = 0, column = 1)

label2 = Label(fen1, text = "Temperature en fahrenheit:")
label2.grid(row = 1, column = 0, sticky = E)

txtCm = Label(fen1, text = "")
txtCm.grid(row = 1, column = 1)

bouton1 = Button(fen1, text = "Convertir", command = convertir)
bouton1.grid(row = 2)

fen1.mainloop()
```

Exercice 2

```
from tkinter import *

def convertir():
    if choix.get() == "C->F":
        fah = float(entree.get()) * 1.8 + 32
        resultat.configure( text = str(fah))
    else:
        celsius = (float(entree.get()) - 32)/1.8
        resultat.configure( text = str(celsius))

def changement(event):
    if choix.get() == "C->F":
        txt1.configure( text = "Temperature en celsius:")
        txt2.configure( text = "Temperature en fahrenheit:")
    else:
        txt2.configure( text = "Temperature en celsius:")
        txt1.configure( text = "Temperature en fahrenheit:")

fen1 = Tk()
fen1.title("Convertir...");

txt1 = Label(fen1, text = "Temperature en celsius:")
txt1.grid(row = 0, column = 0, sticky = E)
```

```

entree = Entry(fen1)
entree.grid(row = 0, column = 1)

txt2 = Label(fen1, text = "Temperature en fahrenheit:")
txt2.grid(row = 1, column = 0, sticky = E)

resultat = Label(fen1, text = "")
resultat.grid(row = 1, column = 1)

bouton = Button(fen1, text = "Convertir", command = convertir)
bouton.grid(row = 2, column = 0)

choix = StringVar(fen1)
choix.set("C->F")
liste = OptionMenu(fen1, choix, "C->F", "F->C", command = changement)
liste.grid(row = 2, column = 1)

fen1.mainloop()

```

Exercise 3

```

from tkinter import *

def tester():
    mot = Entree.get()

    motInv = ""
    for car in mot:
        motInv = car + motInv

    if mot == motInv:
        label2.configure( text = (mot + " est un palindrome"))
    else:
        label2.configure( text = (mot + " n'est pas un palindrome"))

fen1 = Tk()
fen1.title("Palindrome");

label1 = Label(fen1, text = "Mot:")
label1.grid(row = 0, column = 0, sticky = E)

Entree = Entry(fen1)
Entree.grid(row = 0, column = 1)

label2 = Label(fen1, text = "")
label2.grid(row = 1, column = 0, columnspan = 2)

bouton1 = Button(fen1, text = "Tester", command = tester)
bouton1.grid(row = 2, column = 0, columnspan = 2)

fen1.mainloop()

```

Exercice 4

```
from tkinter import *

def afficher():
    nombres = range(20,41)

    for nombre in nombres:
        carres = nombre**2
        cubes = nombre**3
        resultat.insert(INSERT, str(nombre) + "\t" + str(carres) + "\t" +
str(cubes) + "\n")

fen1 = Tk()
fen1.title("Afficher les carres et les cubes");

resultat = Text(fen1)
resultat.config(width=40, height=10)
resultat.insert(INSERT, "Nombre\tCarres\tCubes\n=====\\n")
resultat.grid(row = 0, column = 0, columnspan = 2)

bouton1 = Button(fen1, text = "Afficher", command = afficher)
bouton1.grid(row = 1, column = 0)

bouton2 = Button(fen1, text = "Quitter", command = fen1.destroy)
bouton2.grid(row = 1, column = 1)

fen1.mainloop()
```